

MIXTURE AND ALLIGATION**Key Notes :****Mixture :**

Mixing of two or more qualities together gives the mixture. The quality of the mixture lies between the Least & the Highest of the given qualities.

One variety of quality p_1 of quantity q_1 , second variety of quality p_2 of quantity q_2 , third variety of quality p_3 of quantity q_3 ,.....are mixed together, then the quality of the mixture is given by

$$P = \frac{p_1 \times q_1 + p_2 \times q_2 + p_3 \times q_3 + \dots}{q_1 + q_2 + q_3 + \dots}$$

Alligation :

In allegation, only two qualities are mixed together in a particular ratio to become a mean price.

Alligation Rule :

**Dearer Price
(DP)**

**Cheaper Price
(CP)**

**Mean Price
(MP)**

(MP – CP)

:

(DP – MP)

Note :

Mean value lies between dearer price & Cheaper price

“APTITUDE is Gateway to Career Opportunities” – VENKAT MADAKA**COMPANIES SPECIFIC QUESTIONS – All Models**

1. Sweety purchases two varieties of Apples. One variety of 30 kgs at the rate of Rs.120 /kg and another variety of 20 kgs at the rate of Rs. 150 /kg. Then what is the average price of Apples per kg ?

$$\text{Average price} = \frac{p_1 \times q_1 + p_2 \times q_2}{q_1 + q_2}$$

$$= \frac{120 \times 30 + 150 \times 20}{30 + 20}$$

$$= \frac{3600 + 3000}{30 + 20}$$

$$= \frac{6600}{50} = \text{Rs.132/kg}$$

2. Sujith purchases three varieties of rice. One variety of 20 kgs at the rate of Rs.100/kg, second variety of 30 kgs at the rate of Rs.80/kg, third variety of 50 kgs at the rate of Rs.60/kg. He mixes all these varieties, then what is the average price of rice per kg ?

$$\text{Average price} = \frac{p_1 \times q_1 + p_2 \times q_2 + p_3 \times q_3}{q_1 + q_2 + q_3}$$

$$\text{CAT} = \frac{100 \times 20 + 80 \times 30 + 60 \times 50}{20 + 30 + 50}$$

$$= \text{Rs.74/kg}$$

3. Three varieties of Wheat costing Rs.40/kg, Rs.50/kg and Rs.60/kg are mixed in the ratio 5 : 3 : 2, then what is the average price of wheat per kg ?

Ratio means quantity

$$q_1 : q_2 : q_3 = 5 : 3 : 2$$

$$p_1 = 40, p_2 = 50, p_3 = 60$$

$$\text{Average price} = \frac{p_1 \times q_1 + p_2 \times q_2 + p_3 \times q_3}{q_1 + q_2 + q_3}$$

$$= \frac{40 \times 5 + 50 \times 3 + 60 \times 2}{5 + 3 + 2}$$

$$= \text{Rs.}47/\text{kg}$$

4. Cost of two types of pulses is Rs.15 and Rs, 20 per kg, respectively. If both the pulses are mixed together in the ratio 2:3, then what should be the price of mixed variety of pulses per kg?

$$\text{Average price} = \frac{p_1 \times q_1 + p_2 \times q_2}{q_1 + q_2}$$

$$= \frac{15 \times 2 + 20 \times 3}{2 + 3}$$

$$= \text{Rs.}18/\text{kg}$$

5. Tea worth of Rs. 135/kg &Rs. 126/kg are mixed with a third variety in the ratio 1: 1 : 2. If the mixture is worth Rs. 153 per kg, the price of the third variety per kg will be ____?

$$\frac{P_1 \times q_1 + p_2 \times q_2 + p_3 \times q_3}{q_1 + q_2 + q_3} = 153$$

$$\frac{135 \times 1 + 126 \times 1 + p_3 \times 2}{1 + 1 + 2} = 153$$

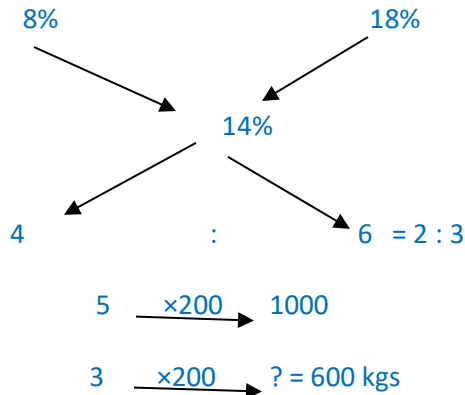
$$\frac{261 + p_3 \times 2}{4} = 153$$

$$P_3 = \text{Rs.}175.5/\text{kg}$$

6. A grocer wishes to sell a mixture of two variety of pulses worth Rs.16 per kg. In what ratio must he mix the pulses to reach this selling price, when cost of one variety of pulses is Rs.14 per kg and the other is Rs.24 per kg?

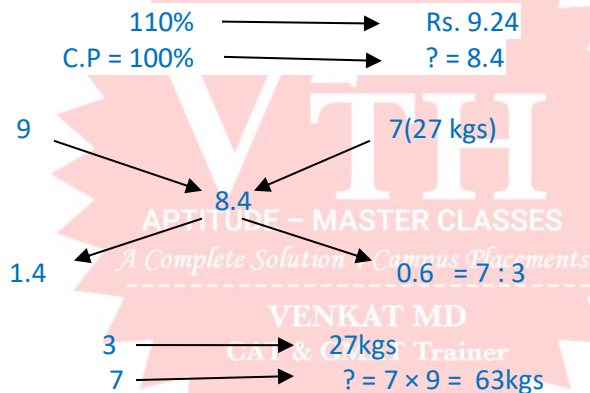
$$\begin{array}{ccc} 14 & & 24 \\ & \searrow \quad \nearrow & \\ & 16 & \\ & \swarrow \quad \searrow & \\ 8 & : & 2 = 4 : 1 \end{array}$$

7. A dealer has 1000 kg sugar and he sells a part of it at 8% profit and the rest of it at 18% profit. The overall profit he earns is 14%. What is the quantity which is sold at 18% profit?

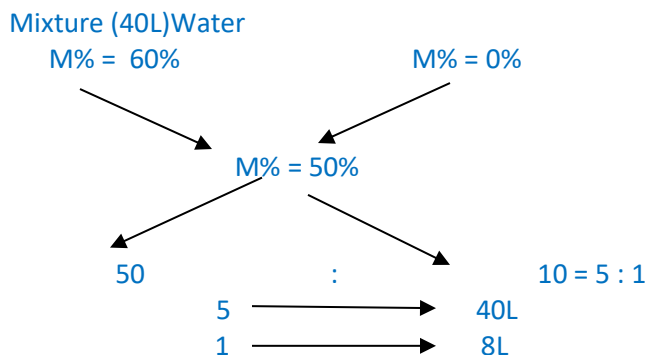


8. How many kilograms of sugar costing Rs. 9 per kg must be mixed with 27 kg of sugar costing Rs. 7 per kg so that there may be a gain of 10 % by selling the mixture at Rs. 9.24 per kg ?

S.P = 9.24, G% = 10%



9. 40L Milk solution has 60% Milk. How many liters of water should be added to the mixture, so that the mixture has 50% Milk ?



10. A person borrowed Rs.10000 from two money lenders. For one he paid 10% for the other he paid 5%. If the total interest for 5 years is Rs.3000, then how much did

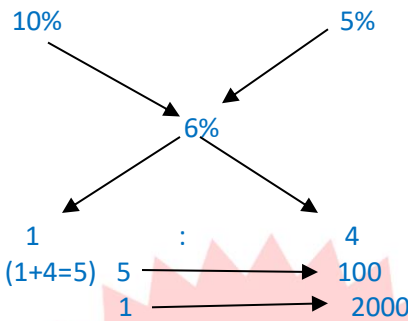
he borrow at 10% rate of interest ?

First, calculate average rate interest.

$$I = \frac{P \times T \times R}{100}$$

$$3000 = \frac{10000 \times 5 \times R}{100}$$

R = 6% (Average rate of interest)



Mixture and Alligation



VENKAT MD
PRACTICE QUESTIONS

1. In 120 liters of a mixture of milk and water in the ratio of 3:1. If a milkman sold 20% of the mixture and added some equal amount of milk and water into the remaining mixture, the ratio of milk and water becomes 11:5 and then find the difference between the amount of milk and water in the final mixture?

- A. 60 liters B. 36 liters C. 48 liters D. 72 liters

2. A vessel contains milk and water in the ratio of 2:1. If 15 liters of mixture is taken out and replaced with water, then the ratio of milk and water becomes 7:5, then find the total quantity of mixture initially?

- A. 90 liters B. 120 liters C. 60 liters D. 150 liters

3. A vessel contains 160 liters of mixture of milk and water, in which the quantity of milk is 200% more than the quantity of water and 40 liters of mixture is removed and x liters of water is added, then the ratio of the milk to water becomes 3:2. Find the value of x.

- A. 30 B. 45 C. 40 D. 50

4. A vessel contains $(x+100)$ liters of milk and x liters of water. If 100 liters of milk and x liters of water is added in the vessel, then the quantity of milk and water becomes equal. Find the quantity of water left in the vessel?

- A.480 liters B.300 liters C.250 liters D.400 liters

5. A vessel contains 100 liters of a mixture of milk and water in the ratio of 3:2. If x liters of mixture is taken out and 25 liters of milk and 20 liters of water are added into the remaining mixture and the quantity of milk in the final mixture is 40% more than that of water, then find the value of x ?

- A.10 B.25 C.15 D.30

6. A vessel contains a mixture of milk and water in the ratio of 2:1. If 30 liters of mixture is taken out and 10 liters of water is added to the mixture, the quantity of milk in the resultant mixture is 30 liters more than that of water, then find the quantity of milk in the vessel finally?

- A.80 liters B.60 liters C.90 liters D.50 liters

7. A vessel contains a mixture of milk and water and the quantity of milk is 80 liters more than that of water. If 60 liters of milk and 80 liters of water are added to the mixture, the ratio of milk to water becomes 3:2, then find the quantity of milk in the vessel initially?

- A.120 liters B.200 liters C.160 liters D.100 liters

8. A vessel contains a certain quantity of milk and x liters of water and 50 liters of mixture is removed in the ratio of 3:2. 40 liters of milk and 40 liters water is added to the mixture, then the ratio of milk to water becomes 4:3. Find the initial quantity of mixture in the vessel.

- A.250 liters B.180 liters C.210 liters D.150 liters

9. A vessel contains 250 liters of a mixture of milk and water. If 150 liters of milk and 100 liters of water are added to the mixture, then the quantity of milk in the resultant mixture is 50% more than that of water. Find the initial quantity of water in the vessel.

- A.80 liters B.50 liters C.75 liters D.100 liters

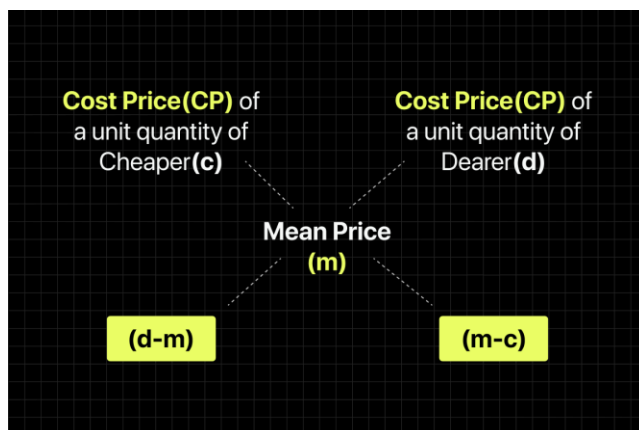
10. A vessel contains a mixture of acid and water in the ratio of 7:2. If 90 liters of mixture is taken out and 20 liters of acid is added to the mixture, then the ratio of acid to water becomes 4:1. Find the initial quantity of mixture in the vessel?

- A.360 liters B.180 liters C.450 liters D.270 liters

V²TH – APTITUDE MASTERY

CAMPUS
PLACEMENT
SKILLS

KEY				
1.C	2.B	3.A	4.D	5.B
6.A	7.A	8.A	9.D	10.D



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----- VENKAT MADAKA

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